Milestone 4 Report

Crowd Monitoring System

1. Abstract

The Crowd Monitoring System is designed to monitor and analyze crowd activity in public places using a camera. The system detects and counts people in different zones and displays live crowd statistics on the screen. It also identifies overcrowding situations and generates alert messages when the crowd exceeds the allowed limit. The system stores crowd data for further analysis and generates reports and graphs to understand crowd patterns. This system helps improve safety and crowd management in areas such as malls, railway stations, and public events.

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2. Introduction

Crowd monitoring is an important aspect of maintaining safety in public environments. Large gatherings of people can lead to overcrowding, which may cause accidents or security issues. Traditional manual monitoring methods are inefficient and time-consuming.

The Crowd Monitoring System uses computer vision techniques to automatically detect and count people in real time. The system analyzes live video

from a camera and monitors the number of people present in different zones. When the crowd exceeds a predefined limit, the system generates alerts. This helps authorities respond quickly and manage crowd situations effectively.

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3. Objectives

The main objectives of the Crowd Monitoring System are:

- Monitor crowd activity in real time
- Detect and count people in different zones
- Identify overcrowding situations
- Store crowd data for analysis
- Generate reports and graphs to understand crowd trends

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4. Tools and Technologies

The system is developed using the following tools and technologies:

- **Python** – Main programming language used for system development
- **OpenCV** – Used for video processing and human detection

- **YOLO Model** – Used for detecting people in video frames
- **NumPy** – Used for numerical computations and data processing
- **CSV Files** – Used to store crowd data such as counts and timestamps
- **Matplotlib** – Used to generate graphs and reports for crowd analysis

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5. System Modules

Store Crowd Data

The system records crowd-related information such as time, zone name, entry count, exit count, and total people present. This data is stored in a CSV file for future analysis.

Live Crowd Statistics

The system displays real-time statistics on the video screen, including the number of people detected in each zone and the total crowd count.

Overcrowding Detection

The system checks whether the number of people in a zone exceeds a predefined limit. If the limit is exceeded, the system identifies the situation as overcrowding.

Alert Message

When overcrowding occurs, the system displays a warning message on the screen to notify the user.

Screenshot Capture

Whenever an alert is triggered, the system automatically captures and saves a screenshot of the camera frame as evidence.

Reports and Graphs

The stored crowd data is used to generate reports and graphs. These visualizations help analyze crowd patterns and understand peak crowd times.

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6. System Architecture

Camera Input

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Video Processing

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Person Detection

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Zone Monitoring

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Crowd Counting

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Overcrowding Detection

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Data Storage (CSV)

The system captures video from a camera, processes each frame, detects people, monitors zones, counts individuals, detects overcrowding, and finally stores the data.

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7. Future Enhancements

The system can be further improved with the following features:

- Mobile alert notifications for security staff
- Cloud storage for crowd data
- Support for multiple cameras
- AI-based crowd prediction and analysis

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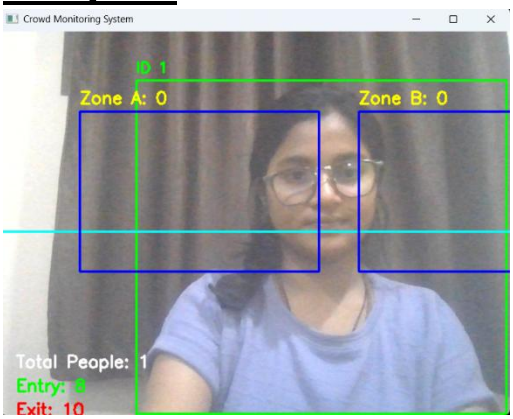
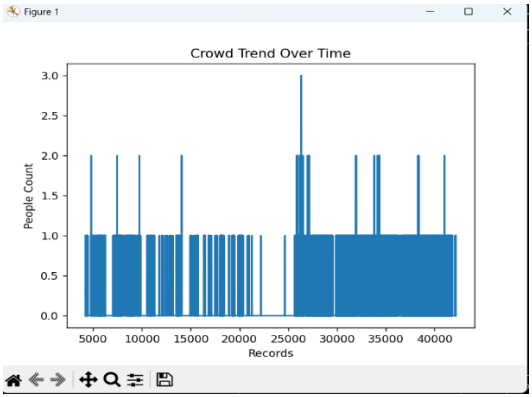
8. Conclusion

The Crowd Monitoring System provides an automated solution for monitoring crowd activity in public spaces. The system detects people, counts them in different zones, and generates alerts during overcrowding situations. It also stores crowd data and produces reports and graphs for analysis. This system can assist authorities in improving safety,

preventing overcrowding, and managing public spaces more efficiently.

Output:




```
Generating Crowd Report...
Total Visitors: 142
Peak Crowd: 3.0

Zone Usage:
Zone
Zone A    3.0
Zone B    2.0
Name: Total, dtype: float64
```